PROJECT REPORT
ON
“Customer Feedback Analysis”

UNIVERSITY INSTITUTE OF COMPUTING (UIC)

SUBMITTED BY:	**SUBMITTED TO:**
Name: Arjun Partap Singh UID: 24BCA10183 Name: Krish Aggarwal UID: 24BCA10184 Section – 24BCA2-A	Name: Mr. Arvinder Singh Subject: Data Science LAB using Excel Subject Code: 24CAP-258

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INTRODUCTION

The Customer Feedback Analysis is a powerful tool developed in Excel to analyse and gain insights from customer feedback data, ultimately aiming to understand and improve customer satisfaction levels.

With the rise of digital platforms such as e-commerce websites, mobile applications, and social media, customers are continuously sharing their experiences. This massive volume of feedback forms a valuable data source, often referred to as "customer voice data." Proper analysis of this data helps organizations make informed decisions and create customer-centric strategies.

Customer feedback can be collected through various methods such as surveys, feedback forms, online reviews, ratings, emails, and direct communication. However, raw feedback alone is not useful unless it is properly processed and analysed. Techniques such as sentiment analysis, keyword extraction, and trend identification are used to convert raw data into meaningful insights.

OBJECTIVES

The main objective of this case study is to understand how customer feedback can be effectively analysed to improve business performance and customer satisfaction. This involves collecting, organizing, and interpreting customer feedback data in a structured manner.

One of the key objectives is to identify customer satisfaction levels. By analysing feedback, businesses can determine whether customers are happy, neutral, or dissatisfied with their products or services. This helps in evaluating overall performance.

Another objective is to identify common issues and problems faced by customers. Feedback often highlights recurring complaints such as poor service quality, product defects, or delayed delivery. Identifying these issues allows organizations to take corrective actions.

The study also aims to understand customer expectations. By analysing feedback trends, businesses can learn what customers value the most, such as quality, price, convenience, or customer support. This helps in designing better products and services.

WHY WE CHOSE THIS TOPIC

I chose the topic "Customer Feedback Analysis" because it is highly relevant in today's digital and customer-focused business environment. Every organization, whether small or large, depends on customer satisfaction for its success. Understanding customer feedback is essential for improving products, services, and overall business strategies.

As a student in the field of computer applications and data science, I am interested in how data can be used to solve real-world problems. Customer feedback provides a rich source of data that can be analysed to generate meaningful insights. This makes it an ideal topic for applying data analysis techniques using tools like Excel.

Another reason for choosing this topic is its practical importance. Businesses across industries rely on customer feedback to improve their offerings. By studying this topic, I can gain knowledge that is directly applicable in real-world scenarios, such as business analytics, marketing, and customer relationship management.

THEORY OF CUSTOMER FEEDBACK ANALYSIS

Customer Feedback Analysis is based on the principles of data analysis, statistics, and customer relationship management. It involves collecting feedback data and transforming it into meaningful insights using various techniques.

- One of the key theoretical concepts is **Sentiment Analysis**, which involves classifying feedback into categories such as positive, negative, and neutral. This helps in understanding customer emotions and satisfaction levels.
- Another important concept is **Data Categorization**, where feedback is grouped based on themes such as product quality, service, pricing, and delivery. This makes it easier to identify specific areas of improvement.
- **Descriptive Statistics** plays a significant role in feedback analysis. Measures such as mean, percentage, and frequency are used to summarize data and identify trends.
- **Text Analysis Techniques** are also used to extract keywords and patterns from textual feedback. This helps in identifying

commonly used words and phrases that indicate customer concerns or preferences.

The theory also includes **Customer Satisfaction Models**, such as the Expectation-Confirmation Theory. According to this theory, customer satisfaction depends on whether the product or service meets or exceeds expectations.

Another concept is **Net Promoter Score (NPS)**, which measures customer loyalty by asking how likely customers are to recommend a product or service.

Data visualization is also an important theoretical aspect. Charts and graphs are used to present data in a clear and understandable format.

Overall, the theory of Customer Feedback Analysis combines statistical methods, data processing techniques, and customer behaviour models to extract valuable insights from feedback data.

IMPLEMENTATION

The implementation of Customer Feedback Analysis involves several steps, starting from data collection to generating insights.

First, feedback data is collected from various sources such as surveys, online reviews, and feedback forms. In this case study, a sample dataset was created in Excel, containing customer ratings, comments, and categories.

Next, the data is organized and cleaned. This includes removing duplicate entries, correcting errors, and ensuring consistency in data format.

After data cleaning, the feedback is categorized based on different aspects such as product quality, service, and pricing. This helps in analysing specific areas.

The next step is sentiment analysis, where feedback is classified as positive, negative, or neutral. This can be done manually or using formulas in Excel.

Statistical analysis is then performed to calculate metrics such as average rating, percentage of positive feedback, and frequency of complaints.

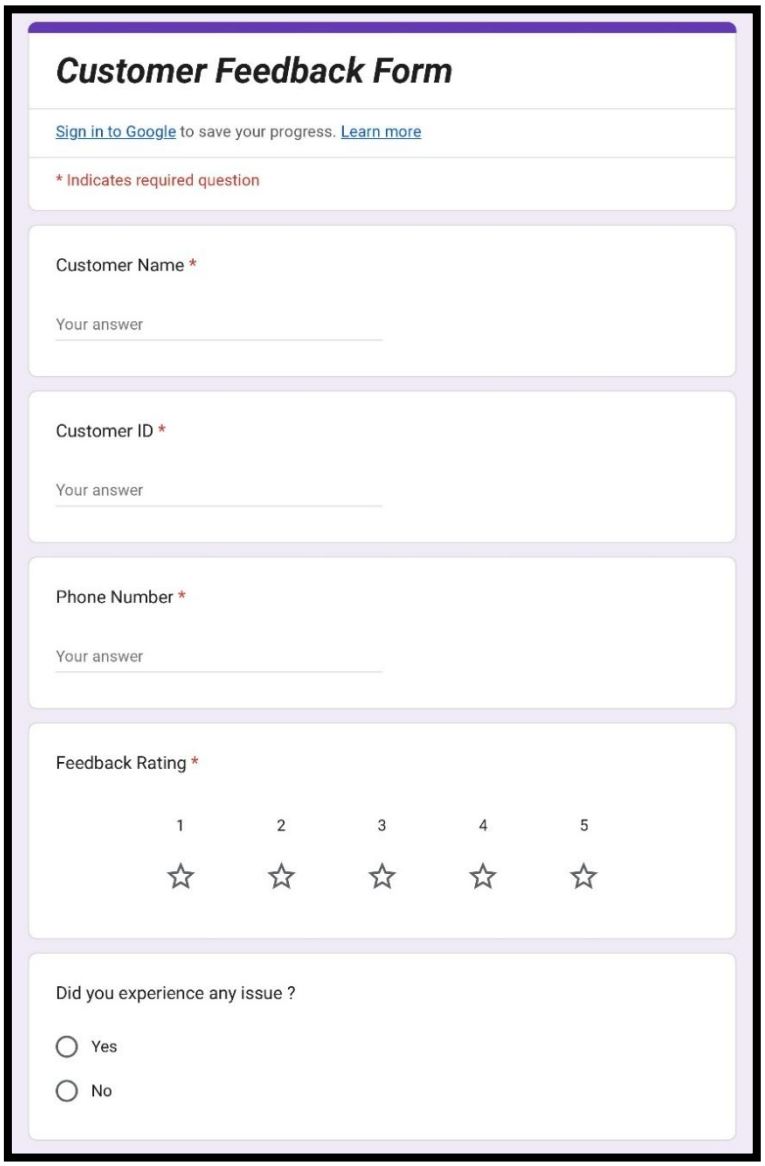
Data visualization techniques such as bar charts and pie charts are used to present the results in a clear and understandable manner.

Finally, insights are generated based on the analysis. These insights help in identifying strengths and weaknesses and suggest areas for improvement.

COMPONENTS USED

❖ GOOGLE FORM:

Google Forms is an effective tool for collecting structured customer feedback in a simple and organized manner. To design a customer feedback form, begin by creating a new form and adding relevant fields to capture essential information.



The first field should be Customer Name, set as a short answer question to identify the respondent. Next, include a Customer ID field, which helps in tracking and linking feedback to specific records in a database or system. This should also be a short answer type and can be validated to accept only specific formats if required.

The third field is Phone Number, which allows follow-up communication if needed. This can be set as a short answer question with response validation to ensure only numeric input of a fixed length (e.g., 10 digits).

For feedback evaluation, add a Rating question using either a multiple-choice or linear scale option ranging from 1 to 5. Here, 1 can represent very poor satisfaction and 5 can represent excellent service. This standardized scale helps in quantitative analysis of customer satisfaction levels.

Finally, include a question on Need for Improvement, formatted as a yes/no multiple-choice question. This provides quick insight into whether customers feel improvements are necessary.

All fields can be marked as required to ensure complete data collection. Once responses are gathered, Google Forms automatically compiles the data into a linked spreadsheet, making it easy to analyse trends, calculate averages, and identify areas for improvement.

❖ *VLOOKUP FUNCTIONALITY:*

VLOOKUP is a powerful Excel function for vertical lookups, searching the first column of a specified table range for a matching value and returning a corresponding value from another column in the same row.

VLOOKUP Syntax:

=VLOOKUP (lookup_value, table_array, col_index_num, [range_lookup]).

- lookup_value: The value to search for (e.g., a customer name in cell A1).
- table_array: The data range, where the first column contains lookup values (e.g., Sheet1!A3:E13); it must start with the lookup column on the left.
- col_index_num: The column number within the table_array from which to retrieve data (1 for first column, 2 for second, etc.).
- range_lookup: FALSE for exact match (recommended to avoid errors) or TRUE for approximate match (requires sorted data).

HOW IT WORKS ?

VLOOKUP scans down the first column of table_array for an exact match to lookup_value. On finding it, it moves right to the specified col_index_num and returns that cell's value. If no match, it returns #N/A. Common pitfalls include unsorted data with TRUE (causing wrong results) or lookup_value not in the first column.

B2 =VLOOKUP(A2,'Raw Data'!A:E,2,0)

	A	B	C	D	E
1	Customer ID	Customer Name	Mobile Number	Feedback Rating (1-5)	Area of Improvement
2	CUST101	Rohan Mehta	9417627782	4	Yes
3	CUST103	Divya Joshi	8745542406	5	No
4					
5					
6					
7					
8					

❖ *CONDITIONAL FORMATTING:*

Conditional Formatting is a powerful feature in Microsoft Excel that helps in identifying trends, patterns, and outliers within a dataset by applying visual cues such as colors, icons, or data bars. It allows users to quickly interpret large amounts of data without manually analyzing each value. This is especially useful in datasets like customer feedback, sales performance, or survey ratings.

For example, in a dataset containing customer ratings, conditional formatting can be applied to visually distinguish between positive, average, and negative feedback. To implement this, select the range of cells containing the ratings and navigate to the "Conditional Formatting" option in the Home tab. From there, choose "New Rule" and apply formatting based on specific conditions.

You can set the following rules:

- Apply a **green fill** to cells where the rating is greater than 5. This indicates high satisfaction or positive feedback.
- Apply a **yellow fill** to cells where the rating is between 2 and 4. This represents average or neutral feedback.
- Apply a **red fill** to cells where the rating is less than 2. This highlights poor feedback or dissatisfaction.

These color codes make it easier to spot trends at a glance. For instance, a dataset dominated by green cells indicates overall positive performance, while more red cells signal areas needing improvement. Additionally, patterns such as repeated yellow or red cells can help identify recurring issues or inconsistencies.

Conditional formatting not only improves data readability but also supports better decision-making. Managers and analysts can use these visual insights to take quick actions, such as addressing customer complaints or improving product quality. Overall, it transforms raw data into meaningful and actionable information.

Customer ID	Customer Name	Mobile Number	Feedback Rating (1-5)	Area of Improvement
CUST101	Rohan Mehta	9417627782	4	Yes
CUST102	Ananya Singh	9499436793	1	No
CUST103	Divya Joshi	8745542406	5	No
CUST104	Aarav Sharma	9787632722	4	Yes
CUST105	Priya Patel	9424141528	1	No
CUST106	Kavya Reddy	9402009065	1	Yes
CUST107	Vikram Nair	9552725414	1	Yes
CUST108	Sneha Iyer	9064479032	3	Yes
CUST109	Rahul Verma	9803588783	5	Yes
CUST110	Arjun Gupta	9773710335	1	No

❖ *Visualization with Graphs and Pie Charts*

Graphs and pie charts turn complex datasets into intuitive visuals, accelerating trend identification and decision-making in performance analysis. Bar graphs excel at categorical comparisons, using bar length to represent metrics like sales volume or ratings—perfect for ranking customers or spotting outliers. Line graphs reveal temporal patterns, showing growth trajectories or seasonal dips over time. Pie charts illustrate part-to-whole relationships, with slices sized by proportion (e.g., 35% dissatisfied customers), ideal when categories total 100% and number under 8 for readability.

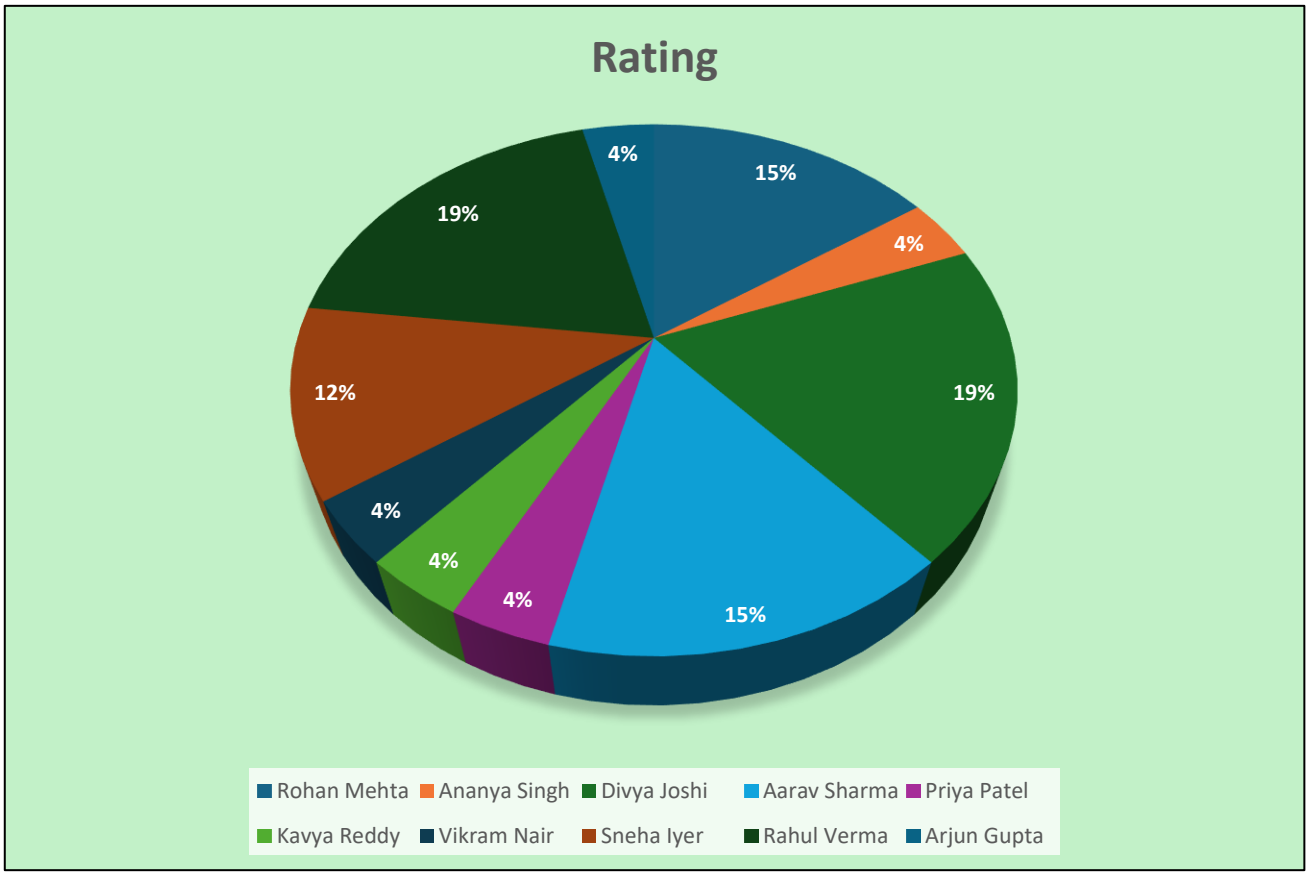
Key principles ensure effectiveness: match chart type to data intent (bar for comparisons, pie for shares), label axes clearly (X: categories, Y: metrics), use consistent colour scales avoiding red-green for accessibility, and minimize clutter with white space.

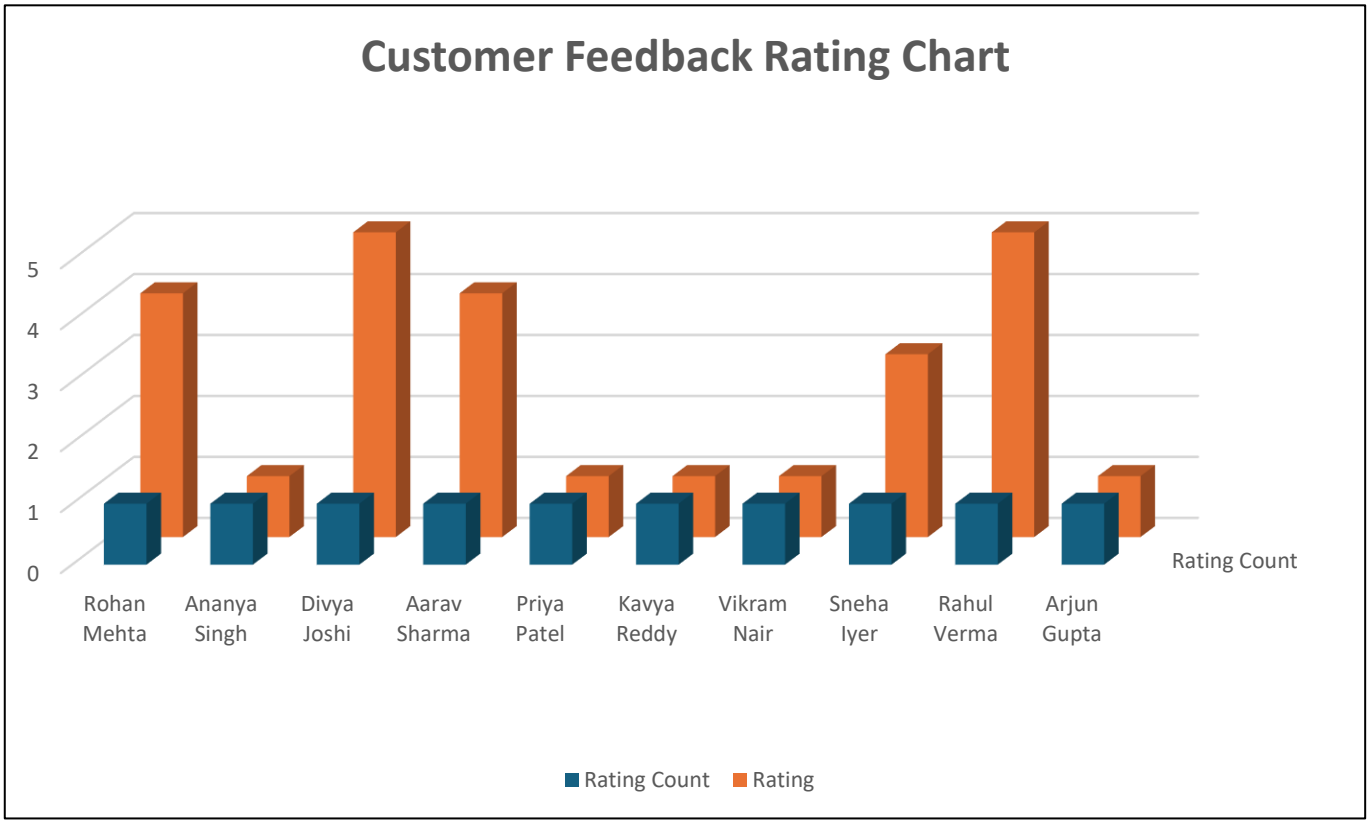
Benefits include instant pattern recognition (top performers, problem areas), stakeholder buy-in through visual storytelling, and exploratory analysis. Excel pivot charts stay dynamic—filter by date/segment, chart updates live. Advanced platforms (Tableau, Power BI) add interactivity: hover details, drill-downs.

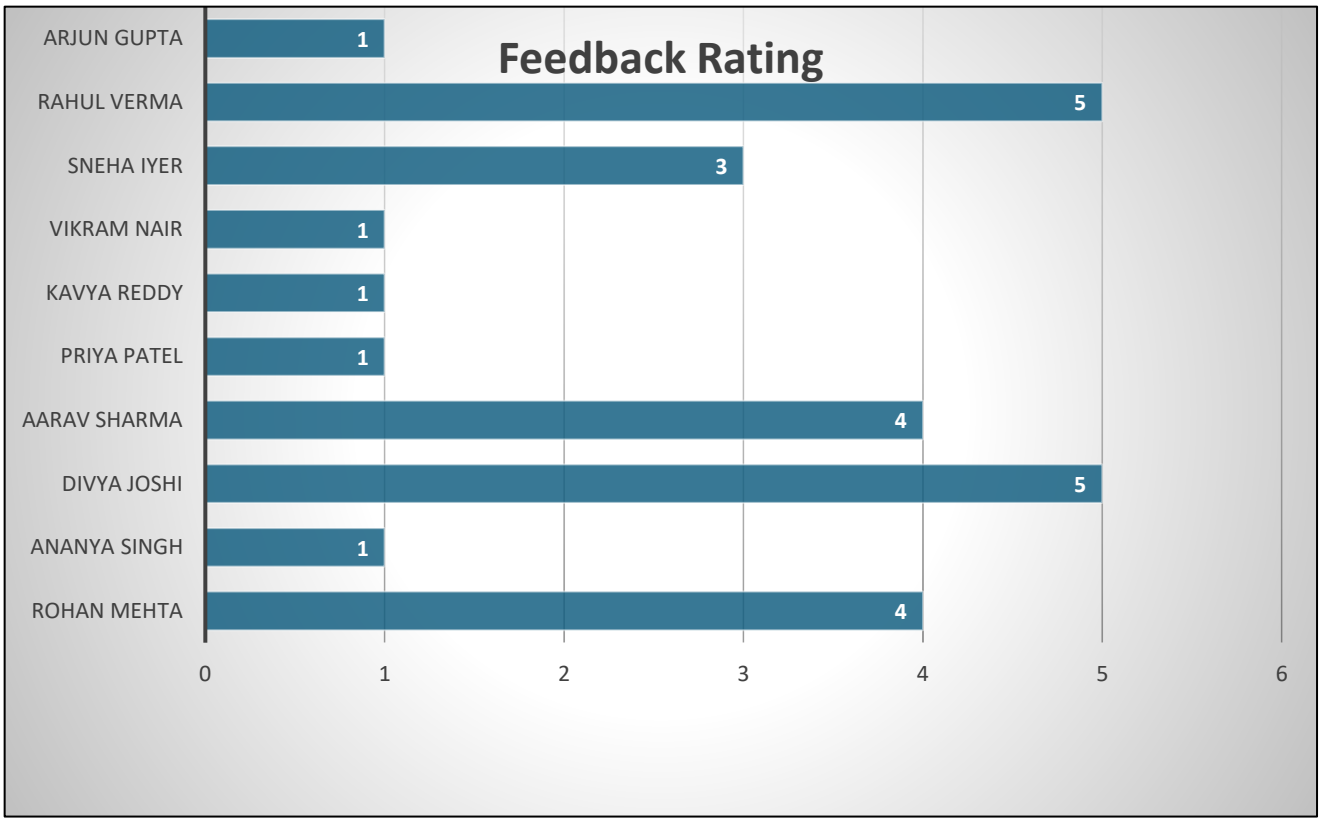
Example Explanation: Bar Graph from Table Data

Step 1: Prepare Pivot Table from customer feedback sheet (CustID, Name, Rating 1-5). Rows: Customer Name. Values: Count of Rating, Average of Rating. Labels show "Aarav (3 responses, 4.2 avg)".

Step 2: Create Chart (Excel): Select pivot → Insert → Clustered Column/Bar. X-axis: Customer labels (Format Axis → Multi-level). Y-axis: Average Rating (1-5 scale); add secondary axis for Count if comparing volume.




Customer Feedback Rating Chart
5
4
3
2
1
0
Rohan Mehta
Ananya Singh
Divya Joshi
Aarav Sharma
Priya Patel
Kavya Reddy
Vikram Nair
Sneha Iyer
Rahul Verma
Arjun Gupta
Rating Count
Rating Count
Rating


Feedback Rating
ARJUN GUPTA 1
RAHUL VERMA 5
SNEHA IYER 3
VIKRAM NAIR 1
KAVYA REDDY 1
PRIYA PATEL 1
AARAV SHARMA 4
DIVYA JOSHI 5
ANANYA SINGH 1
ROHAN MEHTA 4
0
1
2
3
4
5
6

APPLICATIONS IN REAL LIFE

Customer Feedback Analysis using tools like **Google Forms and Microsoft Excel** has wide real-life applications across various industries. In today's digital environment, businesses and organizations actively collect customer opinions through online forms and analyse them to improve their services.

One of the most common applications is in **e-commerce platforms**. Companies collect feedback after every purchase using Google Forms or automated surveys. Customers rate products, delivery services, and overall experience. This data is then exported to Excel, where businesses use charts and graphs to identify trends such as most common complaints or highly rated products.

- In the **education sector**, institutions use Google Forms to collect feedback from students about teachers, courses, and infrastructure. Excel is used to analyse this data through pie charts (for satisfaction levels) and bar graphs (for subject-wise performance). This helps in improving teaching quality and course structure.

- In **restaurants and cafes**, customer feedback forms are used to gather opinions on food quality, service, and ambiance. Excel helps in calculating average ratings and identifying areas needing improvement.
- In the **corporate sector**, companies use employee feedback forms to measure job satisfaction and work environment. HR departments analyse this data using Excel dashboards to improve employee engagement.

IMPLEMENTATION & TESTING

The implementation phase is followed by rigorous testing to ensure accuracy and reliability of the analysis. Initially, the Google Form is tested by submitting sample responses to verify that all questions function correctly and responses are recorded properly in the linked spreadsheet. Any errors in form logic, such as incorrect question branching or missing required fields, are corrected.

Once data is transferred to Excel, the formulas and functions used for analysis are tested. For example, the **AVERAGE()** function is checked by manually calculating sample values to ensure correctness. Similarly, **COUNTIF()** and **IF()** functions are validated against expected outputs. Pivot Tables are reviewed to confirm that they summarize data accurately without duplication or omission.

Charts and graphs are also tested for correctness and clarity. Labels, legends, and axes are verified to ensure that they accurately represent the data. Misleading visualizations are corrected to maintain integrity in reporting.

Testing also includes validating the classification of feedback into positive, neutral, and negative categories. Sample text responses are manually reviewed to confirm that the classification logic is working

as intended. If discrepancies are found, keyword lists or classification criteria are refined.

Performance testing is conducted by increasing the dataset size to ensure Excel can handle large volumes of data without errors or slowdowns. The final report is reviewed for consistency, accuracy, and completeness.

Overall, testing ensures that the system is reliable, accurate, and capable of delivering meaningful insights. It minimizes errors and builds confidence in the results generated from the analysis.

RESULTS AND DISCUSSION

The results of the Customer Feedback Analysis provide valuable insights into customer behavior and satisfaction levels. After processing the data in Excel, key metrics such as average rating, percentage of positive feedback, and most common issues are identified. For instance, the analysis may reveal that 70% of customers are satisfied, while 30% have concerns related to service delays or product quality.

Charts and graphs play a crucial role in presenting these findings. A pie chart may show the distribution of satisfaction levels, while a bar chart highlights the most frequent complaints. Trend analysis using line graphs can indicate whether customer satisfaction is improving or declining over time.

The discussion section interprets these results. High satisfaction scores suggest that the business is performing well in certain areas, such as product quality or customer support. However, recurring negative feedback points to areas that need improvement. For example, if many customers mention delayed responses, it indicates a need to improve customer service efficiency.

The analysis also helps identify patterns and correlations. For instance, lower ratings may be associated with specific products or services, indicating targeted areas for improvement. Positive feedback can be used to reinforce successful strategies and maintain strengths.

Overall, the results provide actionable insights that help organizations make informed decisions. By addressing the identified issues and leveraging strengths, businesses can enhance customer experience and achieve better outcomes.

FUTURE SCOPE

The future scope of Customer Feedback Analysis is vast, especially with advancements in technology. One major improvement is the integration of **automation tools** that can collect and analyze feedback in real time. Instead of manual analysis in Excel, advanced tools and software can process large datasets quickly and efficiently.

Another significant area is the use of **Artificial Intelligence (AI)** and **Natural Language Processing (NLP)**. These technologies can automatically analyze textual feedback, detect sentiment, and identify hidden patterns. This reduces manual effort and increases accuracy in classification.

The project can also be expanded by integrating multiple data sources such as social media reviews, website feedback, and customer support interactions. This provides a more comprehensive view of customer opinions.

Advanced data visualization tools like Power BI or Tableau can be used to create interactive dashboards, allowing businesses to explore data dynamically. Predictive analytics can also be introduced to forecast customer behavior and satisfaction trends.

Additionally, personalization can be improved by analyzing individual customer preferences and tailoring services accordingly. This enhances customer engagement and loyalty.

In the future, real-time feedback systems can be implemented, enabling businesses to respond instantly to customer concerns. This not only improves customer satisfaction but also strengthens brand reputation.

Overall, the project can evolve from a basic Excel-based system to a fully automated, intelligent feedback analysis platform.

CONCLUSION

Customer Feedback Analysis is a powerful tool for understanding customer needs and improving business performance. This project demonstrates how data collected through Google Forms can be effectively analyzed using Excel to generate meaningful insights. By applying formulas, creating charts, and interpreting results, businesses can identify strengths and areas for improvement.

The use of structured data collection methods ensures that feedback is organized and easy to analyze. Excel provides a versatile platform for performing calculations, visualizing data, and summarizing information. The insights gained from this analysis help organizations make informed decisions and enhance customer satisfaction.

One of the key takeaways from this project is the importance of data-driven decision-making. Instead of relying on assumptions, businesses can use actual customer feedback to guide their strategies. This leads to better outcomes and increased customer loyalty.

In conclusion, Customer Feedback Analysis is an essential practice for any organization aiming to succeed in a competitive environment. It not only improves customer experience but also contributes to long-term growth and success.

REFERENCES

1. Microsoft Corporation. *Microsoft Excel Documentation*. Available at: https://support.microsoft.com.
This source provides detailed guidance on Excel functions, formulas, Pivot Tables, and data visualization techniques used in the project.

2. Google LLC. *Google Forms Help Center*. Available at: https://support.google.com/forms
This reference explains how to create forms, collect responses, and integrate them with Google Sheets for data collection.

3. Han, J., Kamber, M., & Pei, J. *Data Mining: Concepts and Techniques*.
A comprehensive book that explains data analysis, pattern recognition, and data preprocessing methods relevant to feedback analysis.

4. Pang, B., & Lee, L. *Opinion Mining and Sentiment Analysis*.
This research paper provides insights into analyzing textual feedback and categorizing sentiment.

GitHub Link:

https://github.com/arjunpsingh45/Data-Science-using-Lab-Project-